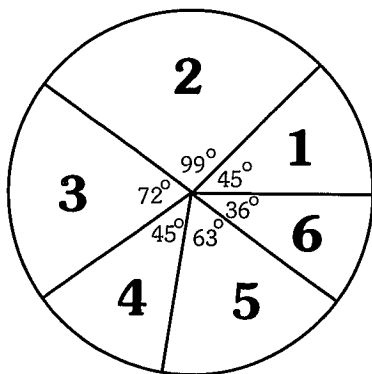


CIRCLE GRAPHS

THROWS OF A DIE



A **circle graph (pie chart)** provides a further alternative when graphing discrete data. It is particularly useful in providing a clear indication of the *relative* contribution of each item of data.

In this case, the 2 is seen to occur most frequently (about one-quarter of the time) and the 6 least often (one-tenth of the time).

The size of each sector of the circle depends upon the *relative frequency* (r.f.) of each item of data. That is, the angle of the sector is found by determining the fraction which the frequency of an item makes with the total frequency and then multiplying it by 360° (full circle).

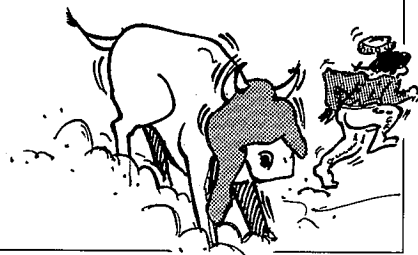
For 1: $\frac{5}{40} \times 360^\circ = 45^\circ$ **For 2:** $\frac{11}{40} \times 360^\circ = 99^\circ$ **For 3:** $\frac{8}{40} \times 360^\circ = 72^\circ$

For 4: $\frac{5}{40} \times 360^\circ = 45^\circ$ **For 5:** $\frac{7}{40} \times 360^\circ = 63^\circ$ **For 6:** $\frac{4}{40} \times 360^\circ = 36^\circ$

(Note: All sector sizes should add to 360°)

ACTIVITY

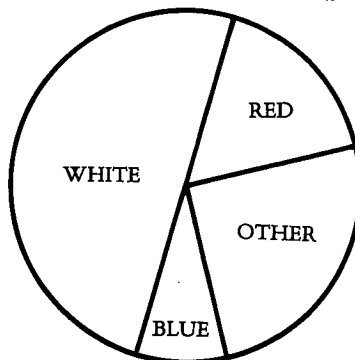
- Survey the members of your class to find their favourite colour.
- Record the results in a frequency table.
- Graph the results with (a) a column graph (b) a circle graph.



EXERCISE 8.2

- Following a traffic survey of car colours, this circle graph was drawn.
 - Measure the angles of the sectors.
 - What percentage of cars were (i) red (ii) white (iii) blue (iv) other?
 - If 120 cars were surveyed altogether, how many were (i) red (ii) white (iii) not blue (iv) red, white or blue?

SURVEY OF CAR COLOURS



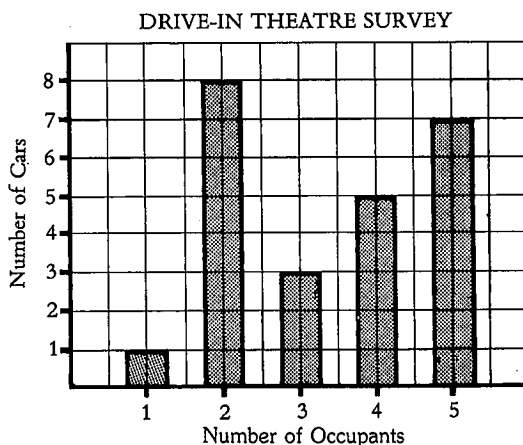
2.

Win	Draw	Loss
7	4	9

At the presentation night for the Marangaroo Soccer Club, the president displayed these figures to show the club's performances throughout the season.

- How many games were played throughout the season?
- What fraction of the games played were (i) won (ii) lost (iii) drawn?
- Draw a circle graph of the data.

3. This column graph represents a sample of the number of occupants per car entering a drive-in theatre from 7.30 p.m. – 7.35 p.m.



- How many (i) cars (ii) people, were involved in the sample?
- Prepare a frequency table of the data.
- What percentage of cars contained:
 - 2 occupants
 - more than 2 occupants?
- Draw a dot frequency diagram of the data.
- Draw a circle graph of the data.

4. The results of a survey to find which sports 50 students played during the weekend are recorded below:

SPORTS PLAYED AT THE WEEKEND

Sports	Tally	Frequency
Hockey		6
Soccer		2
Football		10
Netball		27
Squash		2
Basketball		3
Total		50

- Classify the data as discrete or continuous.
- Construct a circle graph to represent this data.
- What proportion of the sample played:
 - netball
 - squash or basketball?
- Can any conclusions be drawn from this survey? Explain.